DynamiClinic

Switching a multi-agent system's architecture per clinical case: what the prototype does, what it shows, and what it does not yet show.



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github.com/LaansDole/multi-agent-simulation
branch threejs-devall

Clinical multi-agent systems commit to *one architecture* before they see the case

MedAgents

Findings of ACL 2024

Multi-round expert consensus on every question: the same heavy pipeline for an easy case and a hard one.

MDAgents

NeurIPS 2024

Picks solo or team once, by complexity. Best on 7 of 10 benchmarks (up to +4.2%), but the structure cannot change mid-case.

Dialectic-Med

Findings of ACL 2026

Proponent, opponent and mediator falsify a hypothesis; fewer hallucinations on medical image QA.

THE PROPOSAL: MAKE THE TOPOLOGY A VARIABLE

Hierarchical — Routine, low-acuity cases. A hub queries specialists, who do not talk to each other.

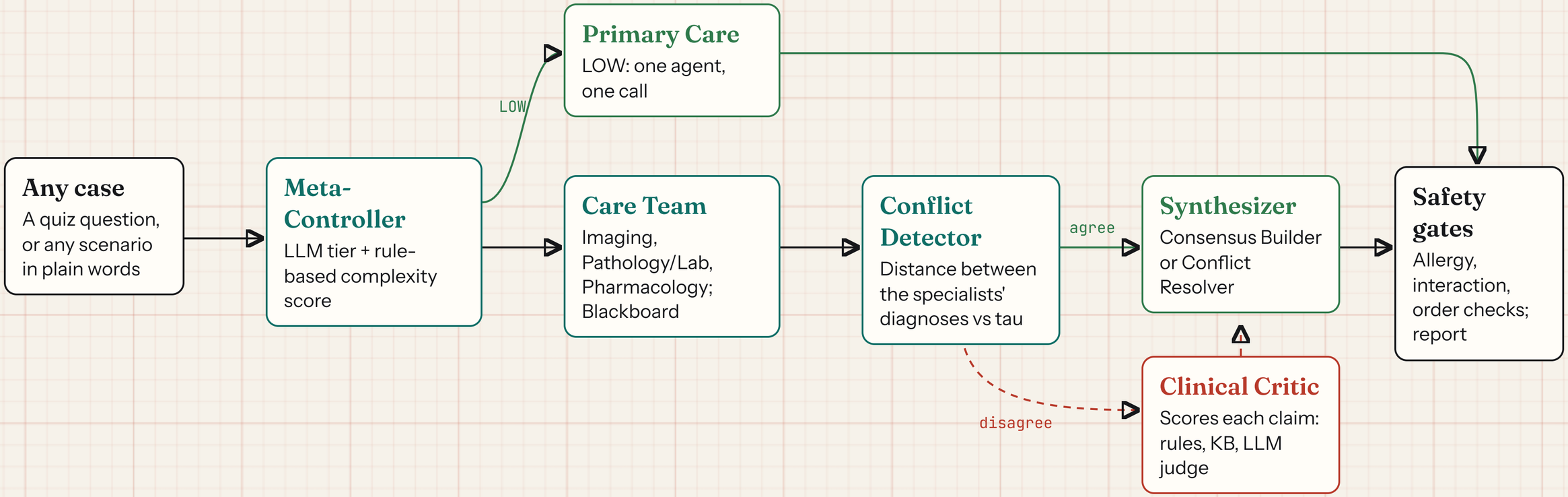
Collaborative — Multi-system cases. Specialists share reasoning and build one differential.

Adversarial — Uncertain or conflicting evidence. A critic attacks the consensus; a mediator resolves.

A **Meta-Controller** scores complexity and uncertainty and switches mode, so heavy compute is spent **only where the case needs it**.

Sources: ACL Anthology 2024.findings-acl.33; arXiv 2404.15155; ACL Anthology 2026.findings-acl.1837

A working pipeline on the DevAll multi-agent platform



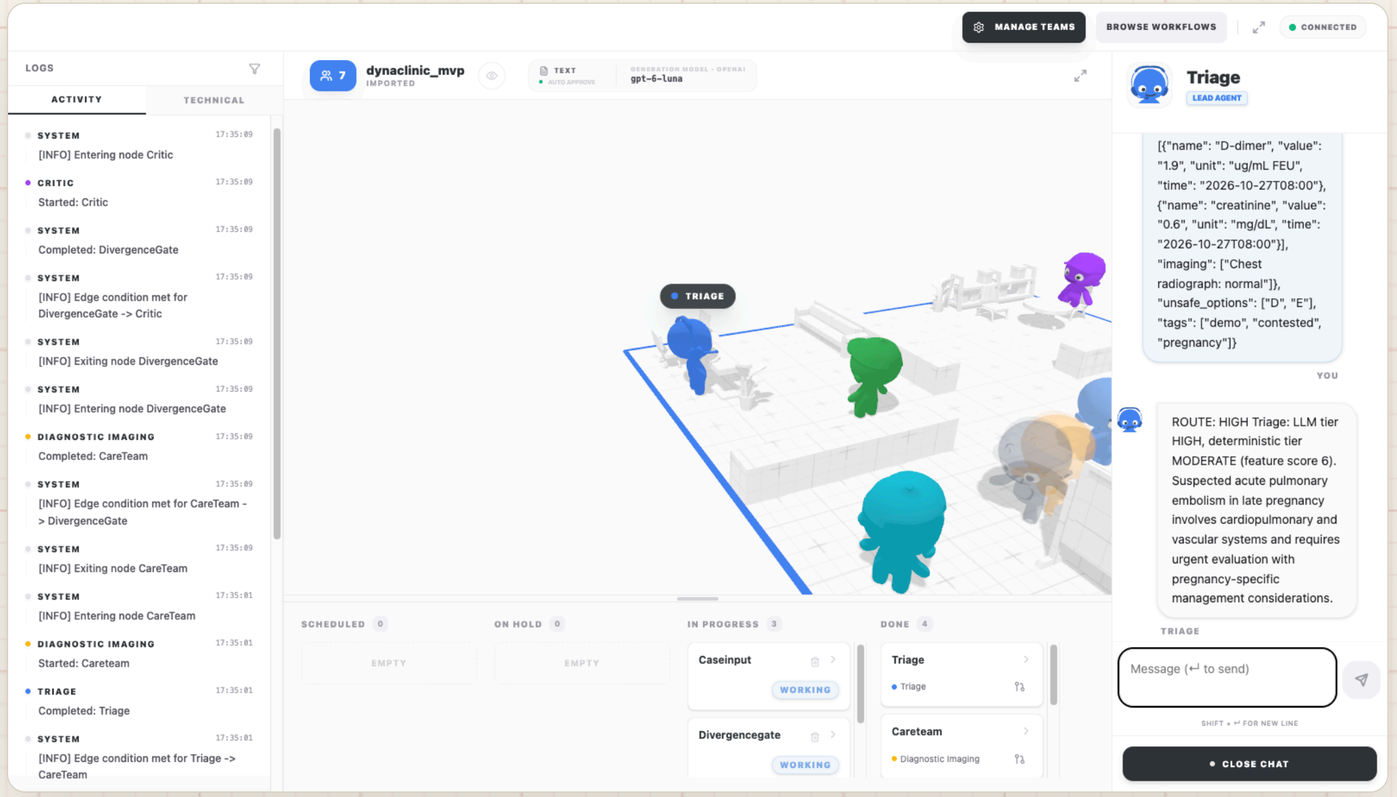
7
custom node types in one YAML workflow

3
arms evaluated: single, dynamic, static (always full team + Critic)

3D
office view: each agent is a character that walks to work

866
automated tests, no model calls needed

The care team disagrees, and the *Critic* is called in



Real run, OpenAI gpt-6-luna. Log: DivergenceGate, then Critic started. Chat: ROUTE: HIGH.

MOCK PATIENT: 30 WEEKS PREGNANT, SUSPECTED CLOT

1. **Triage** calls the case HIGH; all three specialists are consulted.
2. **Imaging** picks B (0.75), **Pharmacology** picks C (0.82), Pathology abstains.
3. **Divergence 0.59 > 0.25**: adversarial mode, the Clinical Critic reviews every claim.
4. The Synthesizer resolves B vs C and answers **C, correct**; safety gates pass.

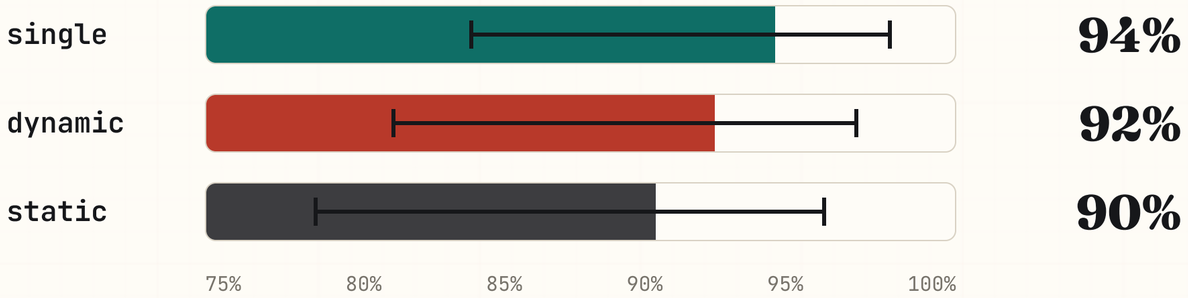
The switch fired in **4 of 9** real runs of this case: all 3 runs triaged HIGH, and 1 of 6 triaged MODERATE. Reported, not forced.

Live demo with video: laansdole.github.io/multi-agent-simulation

Cheaper than an always-on team. *Not more accurate* than one agent, yet.

MedQA-USMLE, 50 random test questions

Accuracy with 95% interval. gpt-6-luna, one run per arm.



	SINGLE	DYNAMIC	STATIC
MedQA calls / question	1.00	3.68	6.00
MedQA latency, mean	5.3 s	19.9 s	29.9 s
MedQA adversarial switch	n/a	0%	always
Synthetic 24 accuracy	100%	100%	100%
Synthetic calls / case	1.00	3.92	6.08
Unsafe answers (synthetic)	0	0	0

Dynamic matches the full team with 36 to 39% fewer calls. The intervals overlap almost completely, and 3 of the 4 dynamic misses were the same wrong answer in every arm: a model-knowledge ceiling, not an orchestration effect.

MedQA: Jin et al. 2021, 4-option test split, seed 2026 sample. Full tables: laansdole.github.io/multi-agent-simulation/results.html

The pipeline matches; the *uncertainty score* and *mesh mode* are still missing

MATCHES

- LOW cases go to one Primary Care Agent
- HIGH cases go to Imaging, Pathology and Pharmacology in parallel, then a Consensus Builder
- Any scenario, not only quiz questions: assessment, plan, red flags, missing information
- Runs on DevAll, as the proposal's Figure 1 shows

PARTIAL

- Complexity score: rule-based, cut-offs not validated
- Conflict Detector: semantic distance between leading diagnoses, as specified; threshold not validated
- Adversarial: one critic round, not proponent and opponent teams
- Today's "collaborative" team is really hub-and-spoke

MISSING

- **Uncertainty score** at triage (entropy of a first generalist pass)
- **Mesh collaboration**: agents never read each other's reasoning
- Counterfactual grounding loss
- One rule over complexity and uncertainty choosing all three modes

PROPOSAL'S OWN EXAMPLE (FIGURES 2, 3)	EXPECTS	ROUTE, 2 RUNS	BEFORE THE FIXES	AFTER: SPECIALISTS' DIAGNOSES
Sore throat and mild fever for 2 days	LOW	LOW, LOW	-	not reached
68-year-old diabetic, chest pain, renal insufficiency, warfarin	HIGH	HIGH, HIGH	fallback, letter answers	all: acute coronary syndrome (0.11, 0.12)
55-year-old, chest pain, normal ECG, high D-dimer, long flight	CONFLICT	MOD, HIGH	fallback, letter answers	all: pulmonary embolism (0.05, 0.14): they agree

Details and run folders: laansdole.github.io/multi-agent-simulation/alignment.html

To test the thesis, the benchmark must make the agents *disagree*

1

A harder benchmark

MedXpertQA (ICML 2025): 4,460 expert-level questions in 17 specialties, filtered to be harder than MedQA. Larger samples to narrow the intervals.

2

Headroom for the team

Repeat with a weaker base model, where a single agent is far from the ceiling and specialists can add value.

3

Finish the controller

Add the uncertainty score and one rule over complexity and uncertainty; build a true mesh mode where agents read each other's reasoning.

4

Safety and validity

Cases with structured data and known unsafe options; thresholds tuned on held-out data; repeated runs; clinician review of the critic's verdicts.

MedXpertQA: arXiv 2501.18362, PMLR v267

What the evidence supports today

1. Switching per case **works end to end** on quiz questions and on any scenario in plain words: triage, team, conflict detection, critic and safety gates run live, and you can watch them in 3D.

2. It **saves compute**: the same accuracy as an always-on team with about a third fewer calls.

3. It does **not yet beat one strong model**, and the adversarial mode is rarely reached: even the proposal's own conflict example produced agreement. The next benchmark has to create disagreement.

`laansdole.github.io/multi-agent-simulation`
Runbook demo: the full journey on video, step by step

`.../results.html and .../alignment.html`
MedQA and synthetic results; the proposal fact check

`github.com/LaansDole/multi-agent-simulation`
Code, eval harness and data, branch threejs-devall

